

Algebra 1 Unit 3 - Part 1 Assessment Guide

General Information	The Unit 3 assessment is divided into two parts. Part 1 assesses benchmarks MA.912.AR.1.2 and MA.912.AR.2.2, and Part 2 assesses MA.912.DP.2.4 and MA.912.DP.2.6. Unit 3 begins by giving students an equation of a line and having the students rewrite the equation in different forms. When a student is just rewriting the equation and not having to use information to create an equation, then the benchmark would be MA.912.AR.1.2. Although students first acquired the skill of rearranging lines into different forms by solving for a given variable, when that rearrangement is part of the process of creating the equation of the line, then the student is considered to be showing knowledge of MA.912.AR.2.2 Writing equations of lines is a fundamental skill of Algebra 1. Most of this assessment assesses MA.912.AR.2.2. Student coherence and fluidity of the forms of lines is a goal of Unit 3. This assessment can be utilized by the teacher as a checkpoint, a formative assessment, a mid-unit assessment, or to assess a student's fluency and coherence with writing and rewriting equations of lines in different forms. This guide is set-up assuming teachers will use Part 1 of the assessment as a mid-unit assessment after Lesson 4. Work is shown in the guide to be helpful to the teacher. Students may not show their work in the same manner. Students should have the freedom to approach writing equations of lines using the given information in whatever way works for them. Some students may want to always use a certain form or students may choose a form that aligns to the given key features. Statewide assessment items may ask students for a certain form of a line, but the student's method of arriving at that form is not assessed. When students have to determine the slope to write the equation of the line, the student may not write the slope in "simplest terms". Each teacher may have different expectations. Therefore, the answer key and guide apply the philosophy that any mathematically correct answer should be accepted. The same holds true for equations written in standard form. The FLDOE defines standard form of the equation of a line as, "$Ax + By = C$, where A, B and C are any rational number." Therefore, it is not the expectation that students write in a form that A, B and C are integer values. Any equivalent equation in the form $Ax + By = C$ should be accepted in this case.
Calculator Information	The assessment is calculator neutral. If students are allowed a calculator, then teachers may wish to guide students to the number of place values to use for decimal equivalents.
Total Recommended Points	The total recommended points in this guide is 96 points. Teachers can either determine where best to use the extra 4 points or teachers may round the total value up to 100 points.

Question	1	Benchmark(s)	MA.912.AR.1.2
$\begin{aligned}8x - 12y &= 48 \\8x - 8x - 12y &= 48 - 8x \\-12y &= -8x + 48 \\\frac{-12y}{-12} &= \frac{-8x}{-12} + \frac{48}{-12} \\y &= \frac{2}{3}x - 4\end{aligned}$		Recommended Scoring (12 Total Points)	Any equivalent equation in slope-intercept form should be accepted as correct. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Slope-Intercept Form	$y = \frac{2}{3}x - 4$		
Question	2a	Benchmark(s)	MA.912.AR.2.2
$m = \frac{-5 - 3}{0 - 2} = \frac{-8}{-2} = 4$		Recommended Scoring (12 Total Points)	Any equivalent equation in point-slope form should be accepted as correct. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Point-Slope Form	$y - 3 = 4(x - 2)$		

Question	2b	Benchmark(s)	MA.912.AR.1.2 MA.912.AR.2.2
$y - 3 = 4(x - 2)$ $y - 3 = 4x - 8$ $y - 3 + 3 - 4x = 4x - 4x - 8 + 3$ $y - 4x = -5$ $-1(y - 4x) = -1(-5)$ Standard Form $4x - y = 5$		Recommended Scoring (12 Total Points)	Any equivalent equation in standard form should be accepted as correct. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. If students have an incorrect equation for question 2a but their equation 2b is equivalent to that incorrect equation, consider giving 5 points. While students showed evidence of MA.912.AR.1.2 by correctly rearranging the equation in question 2a, they did not verify their equation against the graph.

Question	3	Benchmark(s)	MA.912.AR.2.2
Point-Slope Form	$m = \frac{8 + \frac{1}{8}}{-8 - 5} = \frac{\frac{65}{8}}{-13} = -\frac{65}{104}$ $y - 8 = -\frac{65}{104}(x - (-8))$ $y - 8 = -\frac{65}{104}(x + 8)$ $y - 8 = -\frac{5}{8}(x + 8)$	Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Students may write the slope in any equivalent form. Students may choose any point in the table to use to write the equation.
Question	4	Benchmark(s)	MA.912.AR.2.2
Equation	$m = \frac{-11 - (-11)}{3 - (-5)} = \frac{0}{8} = 0$ $y - (-11) = 0(x - 3)$ $y + 11 = 0$ $y + 11 - 11 = 0 - 11$ $y = -11$ $y = -11$	Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Students may write the answer using any form. This should mean that teachers accept equations such as $y - (-11) = 0(x - 3)$ as correct. Students may notice that the y-values are the same and know this means the line is a vertical line and write the equation of the line using only that information.

Question	5	Benchmark(s)	MA.912.AR.2.2
Equation	$y - 15.9 = 4(x - 8.2)$	Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Other forms of the equation are: - Slope-intercept form $y = 4x - 16.9$ - Standard form $4x - y = 16.9$ Any equivalent form should be accepted.
Question	6	Benchmark(s)	MA.912.AR.2.2
Equation	$m = \frac{0 - 6}{-8 - 0} = \frac{-6}{-8} = \frac{3}{4}$ $y - 6 = \frac{3}{4}(x - 0)$ $y - 6 = \frac{3}{4}x$	Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Students may write the answer using any form. Teachers should consider accepting all equivalent form, including forms where the slope of the line is not in simplest terms. Other possible forms of the equation are: - Slope-intercept form $y = \frac{3}{4}x + 6$ - Standard form $3x - 4y = -24$

Question	7	Benchmark(s)	MA.912.AR.2.2
(15, 64) and (26, 112) $m = \frac{112 - 64}{26 - 15} = \frac{48}{11}$		Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider partial credit for minor errors. Other forms of the equation are:
Equation	$y - 64 = \frac{48}{11}(x - 15)$		• Slope-intercept form$y = \frac{48}{11}x - \frac{16}{11}$ • Standard form$48x - 11y = 16$ Any equivalent form should be accepted.